AI Interview Chat Bot

An AI-powered Interview Question Generator built using Streamlit, FastAPI, and Groq LLM (LLaMA 3.3).

Project Overview

This project helps users generate **interview questions and answers** based on topic and difficulty level.

Users can:

- Generate interview questions
- Select difficulty level (Easy / Medium / Hard)
- Choose question type (MCQ / Theory / Coding)
- Get AI-generated answers
- View results in an interactive UI

Technologies Used

Technology	Purpose
Python	Programming Language
Streamlit	Frontend UI
FastAPI	Backend API

| Groq LLM (LLaMA 3.3) | AI Model |

| Requests | API Communication |

| Uvicorn | ASGI Server |

Project Structure

```text
AI-Interview-ChatBot/
│
├── frontend/
│   └── app.py
│
├── backend/
│   └── main.py
│
├── requirements.txt
├── .env
└── README.md
```

Frontend Explanation

The frontend is built using Streamlit.

Frontend Features

- Enter topic for interview
- Select difficulty level
- Choose question type
- Generate AI interview questions
- Expand and view Q&A

Streamlit Components Used

Component	Purpose
st.title()	Page Title
st.text_input()	Input Topic
st.selectbox()	Select Difficulty
st.multiselect()	Select Question Type
st.form_submit_button()	Submit Form
st.expander()	Show Q&A Expandable View
st.write()	Display Output

Backend Explanation

The backend is built using FastAPI and Groq API.

Backend Features

- API endpoint for question generation
- Integration with Groq LLM
- Prompt handling
- Response formatting

API Endpoints

Method	Endpoint	Description
POST	/generate	Generate Interview Questions

API Flow

```text
User Input
  ↓
Streamlit Frontend
  ↓ POST Request
FastAPI Backend
  ↓ Prompt sent to Groq LLM
AI Model Response
  ↓
JSON Output Returned
  ↓
```

Streamlit Displays Q&A

```

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# Installation

## Step 1: Clone Repository

```bash
git clone <repository_link>
cd AI-Interview-ChatBot
```

Step 2: Create Virtual Environment

```bash
python -m venv venv
```

Activate:

Windows

```bash
venv\Scripts\activate
```
```

Mac/Linux

```bash
source venv/bin/activate
```

Step 3: Install Dependencies

```bash
pip install -r requirements.txt
```

Requirements.txt

```txt
streamlit
fastapi
uvicorn
openai
requests
```

Environment Variables (.env)

```env
GROQ_API_KEY=your_groq_api_key_here
```

Run Backend Server

```bash
uvicorn backend.main:app --reload
```

Backend runs at:

```text
http://127.0.0.1:8000
```

Run Frontend Application

```bash
streamlit run frontend/app.py
```

Frontend runs at:

```text
http://localhost:8501
```

Project Workflow

```text
User Input
  ↓
Streamlit UI
  ↓
FastAPI Backend
  ↓
Groq LLM (AI Model)
  ↓
JSON Response
  ↓
Streamlit Display (Q&A)
```

Advantages

- Easy Interview Preparation
- AI-powered learning
- Fast response system
- Beginner-friendly UI

Future Enhancements

- MCQ scoring system
- Timer-based interview mode
- Resume-based question generation
- Save interview history
- Voice-based interview assistant

Author

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License

This project is for educational purposes only.
```
```